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Multi-Source Adaptive-Fusion Transfer Learning for Streamflow Forecasting in Data-Scarce Catchments

Yuxuan Gao[1], **Dongfang Liang**[1], **Edoardo Borgomeo**[1], **and Hiroshi Cho**[2]

[1]Department of Engineering, University of Cambridge, Cambridge, UK, [2]Department of Civil, Environmental and Architectural Engineering, Kumamoto University, Kumamoto, Japan

Key Points:

- A new Transfer Learning (TL) network is proposed to adaptively fuse and transfer knowledge on source catchments to data-scarce catchments
- A small number of carefully selected and high-similarity source catchments can match the transferability of a large number of source data sets
- Across diverse scenarios, the proposed network consistently outperforms fine-tuning TL methods, with larger gains under more severe data scarcity

Correspondence to:
D. Liang,
dl359@cam.ac.uk

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Author Contributions:
Conceptualization: Yuxuan Gao
Formal analysis: Yuxuan Gao
Methodology: Yuxuan Gao
Project administration: Dongfang Liang
Resources: Dongfang Liang
Software: Yuxuan Gao
Supervision: Dongfang Liang, Edoardo Borgomeo
Validation: Yuxuan Gao
Visualization: Yuxuan Gao
Writing – original draft: Yuxuan Gao
Writing – review & editing: Dongfang Liang, Edoardo Borgomeo, Hiroshi Cho

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Abstract Streamflow gauging records remain sparse in many regions, limiting hydrological predictions. Transfer Learning (TL), which uses knowledge from data-rich sources to improve predictions in data-scarce targets, has emerged as a promising solution. Most TL applications in hydrology pre-train deep learning (DL) models on large meteo-hydrological data sets to enhance generalization across diverse catchments. However, this strategy faces two key limitations: (a) TL performance for a specific target may be hindered by irrelevant source inputs and suboptimal fusion of source information; and (b) TL data requirements are demanding. Here, we propose a Multi-Source Adaptive-Fusion Transfer Learning (MSAF-TL) network to address these limitations. MSAF-TL first extracts independent temporal features from each highly related source catchment and then weights them based on the similarity of their static attributes to those of the target. These weighted features are adaptively fused using a cross-attention mechanism, strengthening the final target prediction. We test this framework using a large-sample hydrological data set for Great Britain (CAMELS-GB). Results show that small sets of highly similar source catchments can match the transferability of large data sets. For a target catchment with only 1 year of training data, MSAF-TL achieves an NSE of 0.85 using only the top two similar sources, compared with 0.78 when using all 668 CAMELS-GB catchments. With limited sources, MSAF-TL consistently outperforms conventional multi-source TL (MSTL) across varying source quantities and target conditions. This performance gap widens under greater data scarcity, with average NSE improvements over MSTL reaching 22% when only 6 months of target training data are available. Overall, this study demonstrates that MSAF-TL can achieve more accurate and stable TL performance in data-scarce regions, and that data requirements can be reduced without compromising performance through source pre-selection.

1. Introduction

Accurate hydrological forecasting is essential for flood risk management, water resource management, and evaluating the dual impacts of climate change and human activities on hydrological processes (Alipour & Kibler, 2018). However, global streamflow monitoring networks suffer from significant geographic disparities and uneven record lengths (Do et al., 2018). As of 2025, approximately 11,050 river discharge monitoring stations are operational worldwide, according to the Global Runoff Data Center (GRDC) (2025). However, vast regions, particularly in Africa, South America, and Southeast Asia, remain unmonitored or poorly monitored, with many stations having only 1–10 years of data, greatly limiting their reliability for long-term forecasting and trend analysis (Fekete & Vörösmarty, 2002; Global Runoff Data Centre (GRDC), 2025).

To address the long-standing problem of hydrological prediction in data-scarce (target) catchments, various approaches have been employed and mainly fall into three categories: (a) regionalization approaches based on traditional hydrological models, (b) transfer learning (TL) techniques based on data-driven models, and (c) hybrid modeling approaches that integrate physically based and data-driven models (Lu et al., 2021; Panthi et al., 2021). The third category primarily operates within target catchments themselves. For example, physically based models can be used to generate intermediate variables (e.g., soil moisture, groundwater flow) that serve as inputs to data-driven models, enhancing interpretability while reducing data requirements. Nevertheless, such hybrid approaches encounter difficulties including computational complexity and parameter identifiability challenges (Slater et al., 2025; Xu et al., 2022). Additionally, errors from the physically based component may cascade to the data-driven models, affecting overall model performance (Chen et al., 2023). In contrast, the first two categories focus on knowledge transfer from data-rich (source) catchments to target catchments, which is the core focus of this paper.

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For the first category, traditional hydrological models encompass lumped conceptual models (e.g., GR4J (Charles et al., 2003)), semi-distributed models (e.g., SWAT (Arnold & Fohrer, 2005)), and distributed models (e.g., MIKE SHE (Refshaard & Storm, 1995)). Regionalization approaches derive model parameters for target catchments from the parameters of suitably selected source catchments through various catchment similarity analyses (Pool et al., 2021). However, process-based models inevitably simplify the nonlinear and intricate nature of streamflow generation processes (Xu et al., 2023). Furthermore, the effectiveness of regionalization methods relying on catchment similarity is often constrained by the sparse distribution of nearby gauging stations, which increases catchment heterogeneity and restricts the selection of comparable sources for a given target (Parajka et al., 2015).

For the second category, data-driven models seek to directly learn nonlinear patterns from meteorological and hydrological data (Rahmani et al., 2021). Commonly used deep learning (DL) networks in hydrology include Artificial Neural Networks (ANNs) (Aghelpour & Varshavian, 2020), Convolutional Neural Networks (CNNs) (Ghimire et al., 2021), and Recurrent Neural Networks (RNNs) (Lee & Lee, 2018), along with their variants—Gated Recurrent Units (GRUs) (Le et al., 2021), Long Short-Term Memory (LSTM) (Kratzert et al., 2018), and Transformers (Ghobadi & Kang, 2022). These DL models excel at capturing complex temporal dependencies but typically require large labeled data sets for optimal performance. In data-scarce scenarios, conventional (non-transferable) DL models often suffer from degrading performance or prediction failures (Ma et al., 2021; Yang et al., 2023). TL emerges as a promising solution by transferring knowledge from data-rich source catchments (sources) to data-scarce target catchments (targets) through a pretraining-finetuning paradigm (Pan & Yang, 2009). TL accommodates various applications, including regression (such as streamflow prediction), classification, and clustering tasks (Pan & Yang, 2009). While TL has demonstrated success across numerous fields, including computer vision (Yaqub et al., 2022), natural language processing (Wang & Zheng, 2015), biomedical sciences (Alhares et al., 2023), and finance (He et al., 2020), its adoption in hydrological modeling remains an emerging research frontier.

Several studies have demonstrated the potential of TL for streamflow prediction across diverse regions. Ma et al. (2021) pre-trained an LSTM model on the Catchment Attributes and MEteorology for Large-sample Studies (CAMELS) data set (671 U.S. catchments) and successfully transferred it to Great Britain, Chile, and China, where it outperformed locally trained LSTMs and SWAT-pretrained LSTMs. Khoshkalam et al. (2023) applied LSTM-based TL with data integration to snow-dominated catchments using CAMELS data set, achieving higher accuracy than locally trained LSTMs and physically based HYDROTEL model. Muhammad and Abba (2023) selected 10 relevant source catchments from Model Parameter Estimation Experiment (MOPEX) using Dynamic Time Warping and semantic entropy, and applied a TL + GRU model, which improved streamflow predictions across most sites compared to single TL and TL + LSTM baselines.

Despite recent progress, TL applications in hydrology still face several key challenges, as the transferability of a task is limited by four main factors: large domain divergence, poor source data quality, poor target data quality, and inappropriate TL algorithms and network designs (W. Zhang et al., 2023). Given that data quality issues are often context-specific, the discussion here focuses on domain divergence and network design. Firstly, while pretraining on large-scale data sets enhances model generality across regional data-scarce catchments, it may come at the cost of local performance (Nai et al., 2024). Model performance for a specific target catchment may be constrained when the pre-training data set includes a large number of source catchments with very different rainfall-runoff processes (Nai et al., 2024). It might also become challenging when only a few data-rich catchments surround the target catchment. Secondly, most existing large-sample data sets (multi-source) TL approaches rely on basic LSTM-based models and fuse all source data at the network input level, which may not fully exploit the complementary information across source catchments and also leads to high computational demands (Ma et al., 2021; Yao et al., 2023). Advanced fusion strategies—such as feature-level, output-level, and hybrid fusion—have proven effective in fields like finance (He et al., 2020), energy (Lu et al., 2023), healthcare (Biradar & Virupakshappa, 2024; Jia et al., 2023), mechanical diagnostics (Wu et al., 2023), and image processing (Y. Zhang et al., 2023), yet remain underutilized in hydrology. These techniques offer potential opportunities to improve prediction accuracy under complex hydrological conditions.

In this study, we address this gap by proposing a **M**ulti-**S**ource **A**daptive-**F**usion **T**ransfer **L**earning (MSAF-TL) network. Source catchments are first pre-selected based on weekly hydrograph similarity to identify a small but highly relevant subset of sources for each target. A multi-level adaptive fusion strategy is then applied to integrate

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source knowledge more effectively than conventional input-level fusion, particularly under target data scarcity. The performance of MSAF-TL is benchmarked against three networks: a Non-Transfer Learning (NTL) network (trained only using target data), an All-Source Transfer Learning (ASTL) network (pre-trained on all 668 sources with input-level fusion), and a Multi-Source Transfer Learning (MSTL) network (pre-trained on the selected sources but still using input-level fusion). Three experiments are conducted by varying (a) source catchment quantity, (b) target catchment selection, and (c) length of available target training data. Evaluation metrics include NSE and KGE, together with three flow duration curve based signatures, thereby assessing not only the average fit but also robustness across flow regimes.

The paper is structured as follows: Section 2 introduces the architecture of the proposed MSAF-TL network. Section 3 describes the data sets, source/target catchment selection criteria, and experimental setup. Section 4 presents the experimental results. Section 5 discusses the findings, limitations, and directions for future research. Finally, Section 6 concludes the study.

2. Multi-Source Adaptive-Fusion Transfer Learning (MSAF-TL) Network

2.1. Network Framework

The proposed MSAF-TL network adopts a two-stage training process, as illustrated in Figure 1. In stage 1 (pre-training), multiple LSTM-based Dynamic Feature Extractors (DFEs) (blue branches) are trained independently using dynamic input sequences from each source catchment i (Dyn_S_i, i = 1,2,…,n). These dynamic inputs typically include meteorological time series such as rainfall and temperature. The parameters of each LSTM network are preserved for later use. In stage 2 (tuning), dynamic inputs from the target catchment (Dyn_T) are passed through each pre-trained DFE branch. Users can flexibly determine which LSTM layers to freeze or fine-tune. In parallel, an end-to-end learnable Static Similarity Module (SSM) (pink block) is designed to produce adaptive weights by mapping static attributes into embeddings and then learning a pairwise matching score between the target catchment Sta_T and each source catchment i (Sta_S_i, i = 1,2,…,n). These weights are used to modulate the feature outputs of the corresponding DFE branches, prioritizing more relevant sources.

The weighted features are then integrated using the Adaptive Fusion Module (AFM) (green blocks), which uses a cross-attention (CA) mechanism to integrate information from different sources. Unlike conventional feature fusion — which works by directly concatenating features extracted from multiple source data sets — the AFM first concatenates the features from multiple source catchments and then applies CA between the individual source features and the concatenated feature. This dynamic fusion strategy enables the network to identify key knowledge specific to the target catchment while adaptively integrating complementary information from the sources. Additionally, this fusion mechanism is embedded after each layer of the network, allowing for hierarchical feature integration and progressively enhancing the model's ability to adapt to the target catchment. Further details on the DFE, SSM, and AFM are provided in Sections 2.2–2.4.

2.2. Dynamic Feature Extractor (DFE)

Each DFE branch is a LSTM network, a specialized type of recurrent neural network (RNN) (Hochreiter, 1997). The LSTM network sequentially processes the input feature at each time step, and each LSTM cell generates an output. The output of each cell is conveyed to a fully connected neural network (FCNN) layer to generate the final prediction output. The input and output sequence lengths are the length of the input feature and the final prediction output, respectively. In one single LSTM cell at time step t, the cell state is primarily regulated by three gates: the forget gate, input gate, and output gate. For a detailed explanation of the algorithm of the LSTM cell, readers may refer to previous studies (Greff et al., 2017; Kratzert et al., 2018).

2.3. Static Similarity Module (SSM)

Since DFEs are trained independently on various source catchments, they cannot inherently capture static attribute differences across catchments. Therefore, we introduce an auxiliary SSM, which computes the similarity between each source and the target based on catchment static attributes. Specifically, the SSM consists of multiple static encoders and similarity networks. The static encoders transform static attributes from both source and target catchments into dense embeddings. Let $\text{Sta_S}_i \in \mathbb{R}^{d_s}, i = 1, 2, \ldots, n$ and $\text{Sta_T} \in \mathbb{R}^{d_s}$ denote the static attributes of

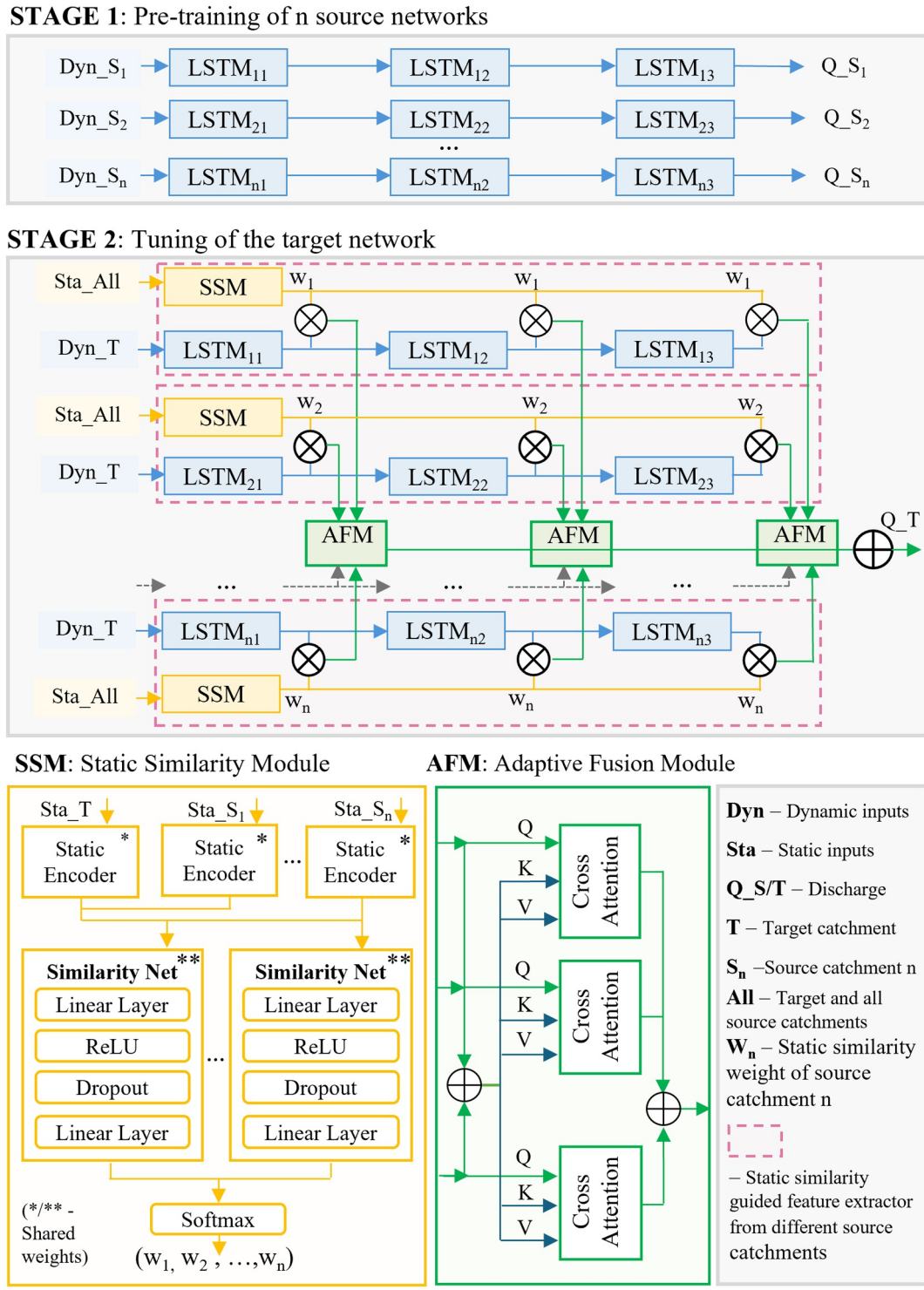


Figure 1. The framework of the Multi-source Adaptive-fusion Transfer Learning (MSAF-TL) network.

the *i*th source catchment and the target catchment, respectively, where d_s is the dimension of static features. The static encoders f_{enc} maps these attributes into dense embeddings:

$$e_S_i = f_{enc}(\text{Sta_S}_i) \in \mathbb{R}^{d_e} \quad (1)$$

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$$e_T = f_{\text{enc}}(\text{Sta_T}) \in \mathbb{R}^{d_e} \tag{2}$$

where d_e is the embedding dimension and f_{enc} is implemented as a multi-layer perceptron with shared weights across all source and target catchments. The shared-weight f_{enc} is designed to make sure all catchments are projected into a common embedding space, making the following similarity scores directly comparable across different sources.

The similarity networks f_{sim} then process these embeddings to produce a similarity score for each source through the Softmax layer:

$$r_S_i = f_{\text{sim}}\left(\left[e_S_i; e_T\right]\right) \tag{3}$$

where $[;]$ denotes concatenation, and $r_S_i \in \mathbb{R}$ is the raw similarity score.

To enhance numerical stability and robustness in similarity calculations, the similarity scores are first constrained within a predefined range to prevent extreme values from adversely affecting the Softmax distribution:

$$\widetilde{r_S_i} = \max\left(-C, \min\left(C, r_S_i\right)\right) \tag{4}$$

where $C > 0$ is the clamping range hyperparameter.

Following this constraint, the scores are scaled by a temperature hyperparameter T to modulate the distribution's sharpness or smoothness:

$$w_S_i = \frac{\exp\left(\widetilde{r_S_i}/\text{T}\right)}{\sum_{j=1}^{n} \exp\left(\widetilde{r_S_j}/\text{T}\right)} \tag{5}$$

where n is the number of source catchments and w_{S_i} represents the normalized similarity weight for source i, satisfying $\sum_{i=1}^{n} w_S_i = 1$. A larger temperature T compresses the logits $\widetilde{r_S_i}/\text{T}$, resulting in a smoother, more uniform distribution in which multiple sources are assigned comparable weights.

Finally, these gating weights are applied to the outputs of each LSTM layer before the cross-attention fusion step. This design enables the model to integrate and transfer both temporal and static knowledge of catchments.

2.4. Adaptive Fusion Module (AFM)

Conventional fine-tuning-based TL in hydrology performs data fusion at the input level (Ma et al., 2021; Yao et al., 2023), overlooking the deep, dynamic relationships and complementary characteristics between features from different sources. To address this limitation, a Cross-Attention (CA)-based AFM is introduced to enhance the integration of the weighted features.

The attention mechanism serves as a fundamental component of the Transformer framework (Vaswani et al., 2017), computing output matrices via Scaled Dot-Product Attention (Equation 6).

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{6}$$

where Q, K, and V are matrices of queries, keys, and values, d_k is the dimension of keys, respectively.

Unlike Self-Attention (SA), where Q, K, and V originate from a single feature space, CA operates on two or more features (Equation 7). In our proposed framework, Q represents the feature of a single source (S_{single}), while K and V are derived from multi-source fused representations (S_{multi}). Through this adaptive attention weighting, the model dynamically learns the interrelationships between heterogeneous sources:

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$$\text{CrossAttention}(S_{\text{single}}, S_{\text{multi}}) = \text{softmax}\left(\frac{Q_{s_{\text{single}}} K_{s_{\text{multi}}}^{T}}{\sqrt{d_k}}\right) V_{s_{\text{multi}}} \tag{7}$$

In practice, we adopt multi-head CA to capture diverse feature subspaces and enhance cross-domain interactions. In this study, we set the number of attention heads to $h = 8$, which can be formulated as Equation 8 (Vaswani et al., 2017). The attention heads enable the model to focus on different representation subspaces and capture richer contextual dependencies. Generally, increasing the number of heads enhances this capability but also demands more computational resources. Based on our preliminary experiments, $h = 8$ achieves satisfactory performance while maintaining efficiency.

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h) W^{O} \tag{8}$$

where $\text{head}_i = \text{Attention}\left(\text{QW}_i^{Q}, \text{KW}_i^{K}, \text{VW}_i^{V}\right)$, and $W^{O} \in \mathbb{R}^{\text{hd}_v \times \text{d}_{\text{model}}}$ in which $d_{\text{model}} = d_k h = d_v h$.

3. Data Preparation and Experimental Settings

Figure 2 presents the workflow of the data preparation and experimental design in this study. The process has four main steps: (a) Data collection, which includes gathering dynamical meteorological inputs and static catchment attributes for the TL networks; (b) Catchment selection, where target catchments are chosen to cover diverse hydrological conditions, and source catchments are chosen to retain the most relevant ones of the target catchment; (c) Experimental settings, which examine the effects of the number of source catchments, the diversity of target catchments, and the length of available target training data; and (d) Model evaluation, which uses multiple performance metrics to assess the TL model prediction ability.

3.1. Data Source

To enable the flexible construction of various data-scarce testing scenarios and a comprehensive evaluation of our proposed network, the CAMELS-GB data set is employed due to its long-term and detailed hydro-climatological records. CAMELS-GB is a large-sample, long-term and open-source data set designed for hydrological modeling across Great Britain (Coxon et al., 2020). Covering the period from 1 October 1970 to 30 September 2015, CAMELS-GB provides daily time series and catchment characteristics for 671 catchments. It includes dynamical meteorological variables such as precipitation, temperature, potential evapotranspiration, radiation, and humidity, as well as hydrological variables such as discharge. In addition to these dynamic records, the data set offers a broad range of static catchment descriptors, encompassing climatic, topographic, geological, hydrometric, anthropogenic, soil, and land use properties.

Based on prior literature (Lees et al., 2021), TL networks in this study use daily precipitation, potential evapotranspiration, and temperature as dynamic inputs, along with a subset of 21 static attributes as auxiliary inputs (detailed in Table 1). The network output is catchment-specific discharge (mm day^{-1}). After assessing data completeness for these 21 static attributes across all 671 catchments, 669 catchments were retained as the final candidate pool of source catchments for subsequent experiments.

3.2. Target Catchment Selection

For the first experiment testing on source catchment quantity, the catchment with a UK National River Flow Archive station ID "39010" (Colne at Denham) was randomly selected as the target catchment to examine the feasibility of the proposed MSAF-TL network (Target 1 in Figure 3). To comprehensively evaluate the generalizability of the MSAF-TL network, three additional target catchments were deliberately chosen from northern, southern, and western regions of the UK. These include: "11003" (Don at Bridge of Alford), "55025" (Llynfi at Three Cocks), and "73014" (Brathay at Jeffy Knotts) (Target 2, 3, and 4 in Figure 3, respectively). Together, the four catchments span a range of climatic, geological, and hydrological conditions across the UK. The locations of the target catchments are shown in Figure 3, and key catchment information is summarized in Table 2.

The first target catchment ("39010") is located in southeastern England's Thames region and is characterized by Chalk with Drift cover geology. The Chilterns scarp is largely rural in the headwaters, but considerable suburban

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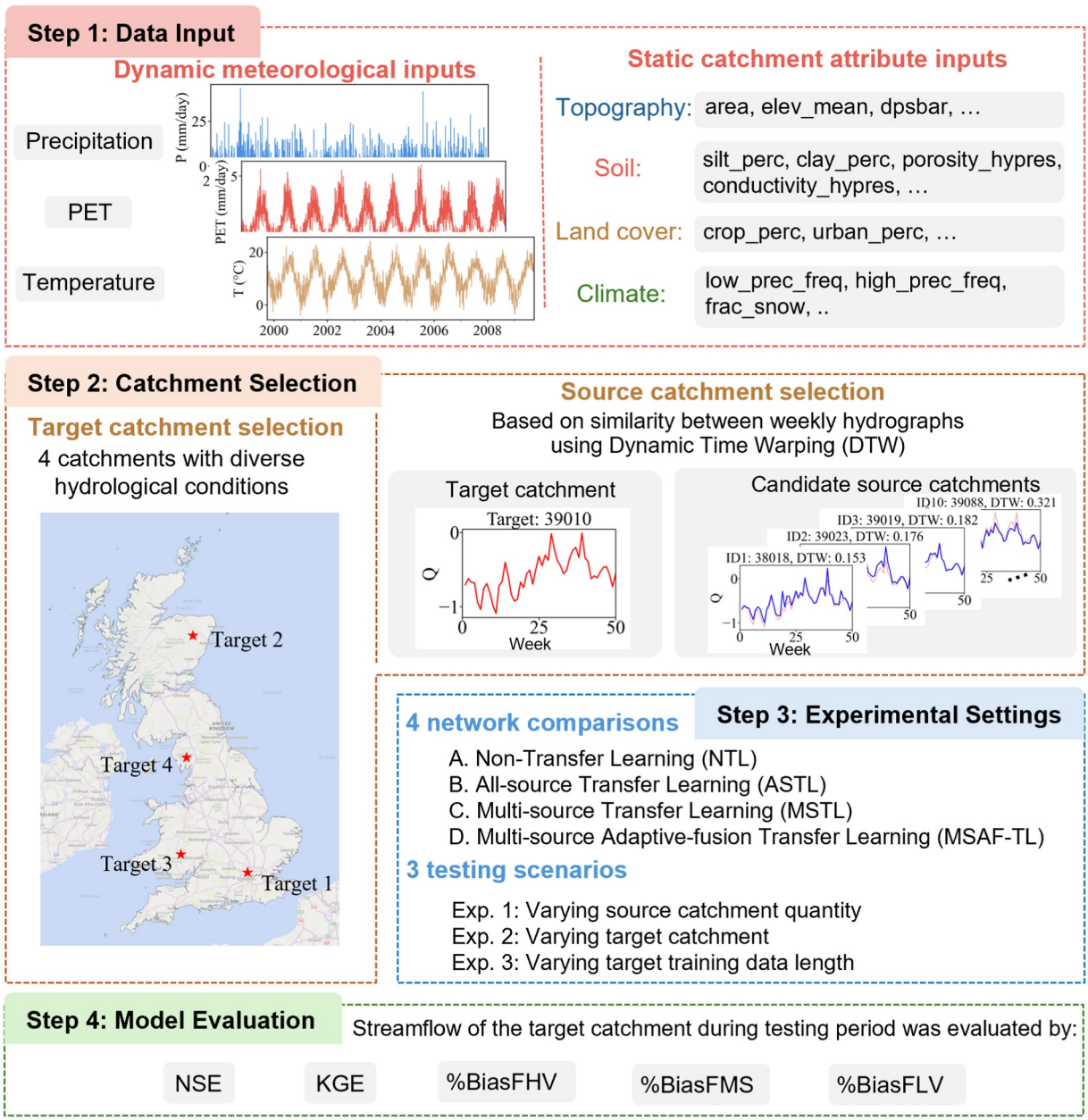


Figure 2. Data preparation and experimental design workflow.

development characterizes the middle and lower reaches. With an average annual rainfall in the standard period (SAAR) of 704 mm, this catchment represents the driest conditions among the selected sites. The second target catchment ("11003") is an upland catchment in northeastern Scotland with mountainous headwaters often experiencing winter snow cover. The Dalradian metamorphic geology supports diverse land uses from high moorland to valley farming. Its moderate SAAR of 967 mm reflects eastern Scotland's intermediate rainfall patterns. The third target catchment ("55025") in Wales features impermeable Old Red Sandstone geology draining the Black Mountains, with Llangorse Lake as a notable feature. The SAAR of 999 mm represents transitional conditions between Wales' wetter western mountains and drier eastern regions. The fourth target catchment ("73014") exemplifies the Lake District's extreme wet conditions, with steep moorland terrain draining into Windermere Lake. Its volcanic geology and exceptionally high SAAR of 2,757 mm make it one of England's wettest locations.

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Table 1
Catchment Dynamic and Static Attribute Inputs From the CAMELS-GB Data Set Used in This Study (Coxon et al., 2020; Lees et al., 2021)

	Description	Unit
Dynamic inputs		
"precipitation"	catchment daily averaged precipitation	mm day^{-1}
"pet"	potential evapotranspiration for a well-watered grass (Penman-Monteith equation)	mm day^{-1}
"temperature"	catchment daily averaged temperature	°C
Static attribute inputs		
"area"	catchment area	km^2
"elev_mean"	catchment mean elevation	m.a.s.l
"dpsbar"	catchment mean drainage path slope	m/km
"sand_perc"	percentage sand—soil	%
"silt_perc"	percentage silt	%
"clay_perc"	percentage clay	%
"porosity_hypres"	volumetric porosity (saturated water content estimated using a pedotransfer function based on silt, clay, and organic fractions, bulk density, and topsoil)	–
"conductivity_hypres"	saturated hydraulic conductivity (estimated using a pedotransfer function based on silt, clay, and organic fractions, bulk density, and topsoil)	cm h^{-1}
"soil_depth_pelletier"	depth to bedrock (maximum 50 m)	m
"frac_snow"	fraction of precipitation falling as snow (for days colder than 0°C)	–
"dwood_perc"	percentage cover of deciduous woodland	%
"ewood_perc"	percentage cover of evergreen woodland	%
"crop_perc"	percentage cover of crops	%
"urban_perc"	percentage cover of suburban and urban	%
"low_prec_freq"	frequency of dry days (<1 mm day^{-1})	days yr^{-1}
"low_prec_dur"	average duration of dry periods (number of consecutive days <1 mm day^{-1})	days
"high_prec_freq"	frequency of high precipitation days (≥5 times the mean daily precipitation)	days yr^{-1}
"high_prec_dur"	average duration of high precipitation events (number of consecutive days ≥5 times the mean daily precipitation)	days
"p_mean"	mean daily precipitation	mm day^{-1}
"pet_mean"	mean daily PET (Penman-Monteith equation without interception correction)	mm day^{-1}
"p_seasonality"	seasonality and timing of precipitation (estimated using sine curves to represent the annual temperature and precipitation cycles; positive (negative) values indicate that precipitation peaks in summer (winter), and values close to zero indicate uniform precipitation throughout the year)	–

For each target catchment, the remaining 668 catchments formed the pool of potential source catchments. The number and selection of source catchments in different sub-experiments varied according to the catchment similarity analysis (see Sections 3.3 and 3.4).

3.3. Source Catchment Selection

Building on previous work in fine-tuning-based TL for data-scarce catchments, we found that using a small number of highly relevant source catchments can match or even outperform the performance of models trained on the full set of metero-hydrological data sets (Gao et al., 2025). To evaluate the potential of MSAF-TL using limited sources while maintaining prediction accuracy, we performed a source catchment pre-selection prior to the comparative experiments.

The pre-selection was based on the similarity of streamflow regimes between each of the 668 source catchments and the target catchment. The similarity was evaluated using shape-based time-series comparison techniques

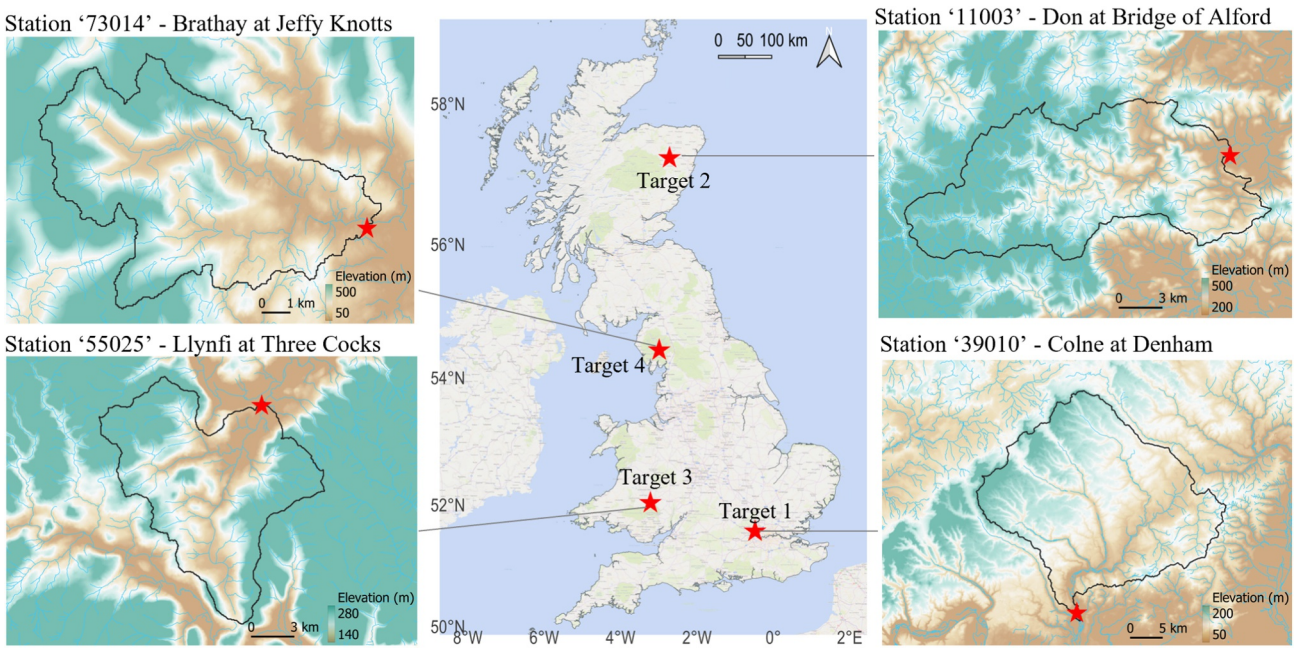


Figure 3. Location and topographic characteristics of the four selected target catchments. Red stars mark gauging-station locations; black outlines indicate catchment boundaries; blue lines show river networks (boundaries and river networks: © UKCEH 2026, contains Ordnance Survey data © Crown copyright and database right; topography: Mapzen Global Terrain).

applied to standardized hydrographs (Yang & Olivera, 2023). Weekly step hydrographs were computed by reshaping daily discharge data into 52-week periods and applying Z-score standardization across catchments to remove magnitude differences, thus emphasizing similarities in temporal flow patterns rather than absolute discharge values. Dynamic Time Warping (DTW) with a Sakoe-Chiba band constraint was employed to measure the distances between source catchments and the target catchment flow regimes, effectively capturing the overall temporal variability of flow series despite minor temporal shifts. Source catchments were then ranked by their DTW distances to the target catchment and used in subsequent experiments.

3.4. Experimental Settings

To systematically evaluate the effectiveness of MSAF-TL, we designed three comparative experiments. These experiments explore the effects of (a) varying the number of source catchments, (b) diverse target catchments, and (c) different lengths of training data from the target catchment. To ensure a fair comparison, all benchmark networks adopt the same three-layer (with 128 neurons) LSTM architecture used in MSAF-TL. The benchmark networks are:

Table 2
Key Information of Selected Target Catchments (Source: UK National River Flow Archive)

Station ID	Location	Drainage area (km^2)	Average annual rainfall in the standard period (SAAR) (1961–1990) (mm)	Primary geology	Dominant land use
"39010"	SE England	743	704	Chalk with Drift cover	Largely rural in the headwaters, suburban development in the middle and lower reaches
"11003"	NE Scotland	499	967	Dalradian metamorphics	Moorland, forestry, hill grazing
"55025"	Wales	132	999	Old Red Sandstone	Grassland with woodland patches
"73014"	Lake District	57.4	2,757	Silurian/Ordovician Volcanics	Steep moorland, natural waterbodies

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Table 3
Detailed Experimental Configurations for MSTL and MSAF-TL Network

Experiment	Pre-training stage				Fine-tuning stage			
	Top n sources	Source catchment training/validating/testing data set length			Selected targets ID	Source catchment training/validating/testing data set length		
Exp. 1: Varying source catchment quantity	Top 2/ Top 6/ Top 10	20-year (30/09/1989–29/09/2009)	3-year (30/09/2009–29/09/2012)	3-year (30/09/2012–29/09/2015)	"39010"	1-year (30/09/2008–29/09/2009)	3-year (30/09/2009–29/09/2012)	3-year (30/09/2012–29/09/2015)
Exp. 2: Varying target catchment	Top 2				"39010"/"11003"/"55025"/"73014"			
Exp. 3: Varying target catchment training data length	Top 2				"39010"/"11003"/"55025"/"73014"	6-month (30/03/2009–29/09/2009)		
						1-year (30/09/2008–29/09/2009)		
						2-year (30/09/2007–29/09/2009)		

1. Non-Transfer Learning (NTL) network: Trains the LSTM network exclusively on limited target catchment data without incorporating any TL techniques. Since only a single target catchment is involved, this network processes only dynamic data without considering static catchment attributes.
2. All-Source Transfer Learning (ASTL) network: Pre-trains the LSTM network on all 668 source catchments, then applies conventional fine-tuning strategies on target catchment data (Ma et al., 2021). Both dynamic and static data from all 668 catchments are fused at the input layer and fed into a single LSTM network for pre-training.
3. Multi-Source Transfer Learning (MSTL) network: According to the flow regime similarity analysis, a small subset of top-ranked source catchments is selected. The LSTM network is pre-trained on the selected sources, then applies conventional fine-tuning strategies on target catchment data. Both dynamic and static data from the selected source catchments are fused at the input layer and fed into a single LSTM network.

Table 3 summarizes the experimental configurations for the MSTL and MSAF-TL networks, including the selected source and target catchments, and the corresponding training/validating/testing periods for both the pre-training and fine-tuning stages across three experiments.

3.5. Model Training

Model training was primarily conducted on a local workstation with an Intel® Core™ i9-14900KF processor and an NVIDIA® GeForce RTX™ 4080 SUPER GPU. Additional training was accelerated using containerized instances on the Beijing Super Cloud Computing Center, equipped with NVIDIA® RTX™ 4090 GPUs and 10 virtual CPU cores. The open-source Neuralhydrology codebase (written in Python) was used as the foundational framework of TL experiments, available at: https://github.com/neuralhydrology/neuralhydrology (Kratzert et al., 2022). The proposed MSAF-TL network code is available at https://github.com/YuxuanGaoYG/MSAF-TL.

For the pre-training stage of MSAF-TL and all benchmark networks, the "cudalstm" architecture from the Neuralhydrology codebase was used. Hyperparameters were tuned via grid search over the following configurations: batch size ([32, 64]), sequence length ([90, 180, 365]), initial learning rate ([10^{-3}, 10^{-4}]), and output dropout ([0.4, 0.6]). The hidden size and number of LSTM layers are fixed at 128 and 3, respectively, to ensure compatibility in the tuning step using the MSAF-TL network. Each model was trained for 50 epochs with an early stopping mechanism. The loss function was mean square error (MSE), which measures the regression loss between predicted and observed flows. The model with the best validation performance on Nash-Sutcliffe Efficiency (NSE) value was selected as the optimal model, and the weights and bias were saved for the following TL experiments.

For the tuning stage of MSAF-TL, ASTL, and MSTL networks, grid search was again applied for batch size ([32, 64]), initial learning rate ([10^{-3}, 10^{-4}]), and output dropout ([0.4, 0.6]). Specially for the MSAF-TL network, three more configurations of static similarity dropout ([0.4, 0.6]), clamp range ([10, 20]) and temperature ([1.5,

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2.0]) were added. The clamp range and temperature were set for the similarity scores constraining and sharpness reducing in the SSM, respectively (as discussed in Section 2.3). The tuning step was run for up to 100 epochs with an early stopping mechanism.

Additionally, three fine-tuning strategies were adopted for every TL-based network: (a) fine-tuning only the linear output layer, (b) fine-tuning the third LSTM layer and linear layer, and (c) fine-tuning the second and third LSTM layers along with the linear layer. This setup enables a controlled comparison between minimal adaptation and deeper representational adaptation, while still retaining the most general temporal patterns learned during pre-training. The one with the best validation performance (based on NSE) was used for final testing. The optimal values of each hyperparameter for every experiment are provided in Appendix B.

3.6. Evaluation Metrics

Model evaluation during testing periods was assessed using several evaluation metrics, including Nash-Sutcliffe Efficiency (NSE), Kling-Gupta Efficiency (KGE), and three flow duration curve (FDC)-based signature indicators: %BiasFHV, %BiasFMS, and %BiasFLV. NSE and KGE are widely used hydrological metrics with values closer to 1 signify better predictive accuracy (Equations 9 and 10). The FDC-based indicators quantify biases across different flow regime segments, focusing on high-flows (top 2% of flow values, %BiasFHV), mid-range flows (20th–70th percentiles; %BiasFMS), and low flows (exceedance > 70%; %BiasFLV) (Equations 11–13).

$$\mathrm{NSE} = 1 - \frac{\sum_{t=1}^{n} (\mathrm{QO}^t - \mathrm{QP}^t)^2}{\sum_{t=1}^{n} (\mathrm{QO}^t - \overline{\mathrm{QO}})^2} \tag{9}$$

where n is the number of observations, QO^t is the observed flow at time t, QP^t is the predicted flow at time t, $\overline{\mathrm{QO}}$ is the mean value of observed flows (Nash & Sutcliffe, 1970).

$$\mathrm{KGE} = 1 - \sqrt{(r-1)^2 + \left(\frac{\sigma_\mathrm{P}}{\sigma_\mathrm{O}} - 1\right)^2 + \left(\frac{\mu_\mathrm{P}}{\mu_\mathrm{O}} - 1\right)^2} \tag{10}$$

where r is the linear correlation coefficient between observed and predicted flows, σ_p and σ_o are the standard deviations of predicted flows and observed flows, μ_p and μ_o are the mean values of predicted flows and observed flows (Knoben et al., 2019).

$$\%\ \mathrm{BiasFHV} = \frac{\sum_{h=1}^{\mathrm{H}} (\mathrm{QP}_\mathrm{h} - \mathrm{QO}_\mathrm{h})}{\sum_{h=1}^{\mathrm{H}} \mathrm{QO}_\mathrm{h}} \times 100 \tag{11}$$

where $h = 1, 2, \ldots, \mathrm{H}$ denotes the indices for high flows with exceedance probabilities lower than 0.02 (Yilmaz et al., 2008).

$$\%\ \mathrm{BiasFMS} = \frac{[\log(\mathrm{QP}_{\mathrm{m1}}) - \log(\mathrm{QP}_{\mathrm{m2}})] - [\log(\mathrm{QO}_{\mathrm{m1}}) - \log(\mathrm{QO}_{\mathrm{m2}})]}{[\log(\mathrm{QO}_{\mathrm{m1}}) - \log(\mathrm{QO}_{\mathrm{m2}})]} \times 100 \tag{12}$$

where m1 and m2 are the lowest and highest flow exceedance probabilities of 0.2 and 0.7, defining the mid-segment of the FDC (Yilmaz et al., 2008).

$$\%\ \mathrm{BiasFLV} = -1 \cdot \frac{\sum_{l=1}^{\mathrm{L}} [\log(\mathrm{QP}_\mathrm{l}) - \log(\mathrm{QP}_\mathrm{L})] - \sum_{l=1}^{\mathrm{L}} [\log(\mathrm{QO}_\mathrm{l}) - \log(\mathrm{QO}_\mathrm{L})]}{\sum_{l=1}^{\mathrm{L}} [\log(\mathrm{QO}_\mathrm{l}) - \log(\mathrm{QO}_\mathrm{L})]} \times 100 \tag{13}$$

where $l = 1, 2, \ldots, L$ refers to the indices of flow values within the low-flow segment of the FDC, corresponding to exceedance probabilities ranging from 0.7 to 1.0, with L denoting the index of the minimum flow (Yilmaz et al., 2008).

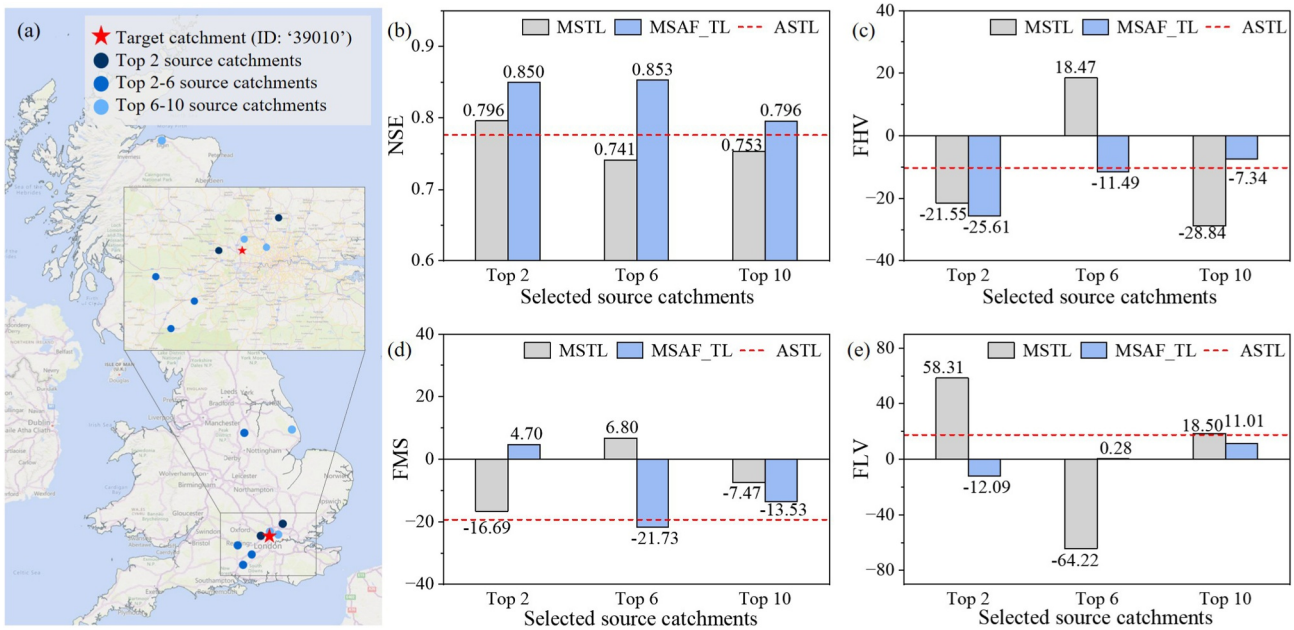


Figure 4. Performance comparison of three TL networks using different numbers of source catchments. (a) Spatial distribution of target catchment (ID: "39010") and selected source catchments. (b–e) NSE, %BiasFHV, %BiasFMS, and %BiasFLV results for MSTL, MSAF-TL, and ASTL across three source catchment groups.

4. Results

4.1. TL Performance Evaluation With Varying Source Catchment Quantities

This section investigates how the number of similar source catchments affects TL performance for the target catchment (ID: "39010"). Notably, the non-transfer learning (NTL) network, trained solely on 1 year of target data, performs poorly, with an NSE of 0.148, %BiasFHV of −55.40%, %BiasFMS of −36.50%, and %BiasFLV of 88.57%. These results indicate that such limited data is insufficient for securing reliable predictions. Therefore, NTL results are excluded from Figure 4 to maintain clarity and focus on TL-based comparisons.

Figure 4a shows the spatial distribution of the target and selected source catchments, grouped by similarity rank: top 2, top 6, and top 10. Detailed comparisons of flow regimes and corresponding DTW values are provided in Appendix A. Figures 4b–4d present the evaluation results—NSE, %BiasFHV, %BiasFMS, and %BiasFLV—for the MSTL, MSAF-TL, and ASTL networks.

Figure 4b shows that MSTL achieves NSEs of 0.796, 0.741, 0.753 for the top 2, 6, and 10 source configurations, respectively, while MSAF-TL achieves NSEs of 0.850, 0.853, 0.796 in the same configurations. Both networks perform comparably to ASTL (NSE = 0.776) despite using only 10 or fewer source catchments, demonstrating that selecting sources based on flow regime similarity effectively reduces data requirements without sacrificing predictive performance for this target catchment. Moreover, MSAF-TL consistently outperforms MSTL, with NSE improvements of 6.78%, 15.02%, and 5.72% for the three source configurations respectively. These results highlight the advantage of the proposed adaptive fusion strategy in improving transferability under various limited-source scenarios.

In terms of high-flow prediction, Figure 4c shows the %BiasFHV values, which quantify biases in the top 2% of flow values with negative values indicating underestimation. MSAF-TL exhibits a clear reduction in %BiasFHV as more source catchments are added—improving from −15.23% (top 2) to −7.34% (top 10)—even outperforming ASTL (−11.49%). In contrast, MSTL observes no significant improvement in prediction accuracy. This finding potentially indicates that the proposed network can better integrate the high-flow knowledge from a broader range of extreme events. In mid-flow prediction (Figure 4d) which reflect biases in 20th–70th percentiles flows, MSAF-TL does not show consistent advantages over MSTL across different source configurations. However, across most configurations, both networks outperform ASTL (%BiasFMS = −21.73%).

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The most notable advantage of MSAF-TL emerges in low-flow predictions (flows with exceedance over 70%) (Figure 4e). MSAF-TL achieves %BiasFLV values of −12.09%, 0.28%, and 11.01% for top 2, 6, and 10 configurations, respectively, significantly better than MSTL and ASTL. This result underscores the strength of the adaptive fusion mechanism in capturing low-baseflow characteristics.

4.2. TL Performance Evaluation Across Diverse Target Catchments

To further validate the generalizability of the proposed network, this section compares the MSAF-TL with other networks on 3 additional target catchments. Among the networks, the MSTL and MSAF-TL use only the top 2 source catchments, while ASTL uses all the 668 source catchments. The spatial distributions of each target catchment (red star) along with their corresponding top 2 source catchments (blue circles) are shown in Figures 5b, 5d, 5f, and 5h. For reference, the NSE values of the NTL network for the three target catchments (IDs: "11003", "55025", and "73014") are 0.403, 0.142, and 0.017, respectively. These results are significantly lower than those achieved by the TL-based methods, so NTL results are excluded from Figure 5 to maintain visual clarity.

Figure 5a presents the NSE performance of MSAF-TL against MSTL and ASTL across four target catchments. MSAF-TL consistently outperforms MSTL, achieving NSE improvements of 6.78%, 10.41%, 7.38%, and 4.19%, respectively. The corresponding discharge curves in Figures 5c, 5e, 5g, and 5i further visualize that for the three newly added target catchments, MSAF-TL provides closer alignment with observed discharges than MSTL, particularly during high-flow periods. The %BiasFHV improves from −19.27% to −10.50%, −32.40% to −31.15%, and −15.00% to −14.71% for the three target catchments, respectively. Full evaluation metrics are available in Appendix C. These results demonstrate that when only limited source catchments are available, our proposed network exhibits more robust and effective transferability across diverse target catchments.

Beyond the overall improvements, the results also reveal differences in simulation performance across target catchments. These differences arise from two aspects: (a) the variation of MSTL performance across targets, and (b) the variation in the improvements achieved by MSAF-TL over MSTL. Regarding the differences of MSTL performance, the results show an interesting contrast. Target "11003" has the best NTL performance and Target "73014" has the worst; however, under MSTL, Target "11003" becomes the worst case while Target "73014" becomes the best one. Two reasons may explain this phenomenon. First, the amount of useful information provided by the source catchments may differ from one target to another, leading to unequal transfer benefits. Second, MSTL may fail to properly balance target-specific knowledge and source-provided knowledge. When the target already contains relatively rich information, the model may be disturbed by transferred signals, thereby hindering effective selection of source knowledge.

For the differences in improvements brought by MSAF-TL over MSTL, each target catchment uses the same set of selected sources, which removes the effect of varying source information and makes the observed gains attributable to the network design itself. In this case, Target "11003" exhibits the largest improvement while Target "73014" shows the smallest improvement, suggesting that MSAF-TL can more effectively exploit source knowledge to enhance TL performance (further discussed in Section 5.2).

In addition to outperforming ASTL on the previously discussed target 1 ("39010"), MSAF-TL also achieves better performance on target 4 ("73014"). Although it does not surpass ASTL on the remaining two target catchments, it still achieves satisfying NSE values of 0.796 and 0.832 by using only 2 source catchments. This underscores the competitive performance of the proposed strategy, implying potential to reduce the data requirements in large-scale hydrological TL applications.

4.3. TL Performance Evaluation Across Various Data Lengths From Target Catchments

This section investigates how the length of training data from the target catchment affects the performance of MSTL and MSAF-TL across four catchments. Figure 6 presents the NSE results when networks are trained with 2-year, 1-year, and 6-month data from each target catchment. Results for the complete evaluation metrics are provided in Appendix C.

Across all four targets, MSAF-TL consistently outperforms MSTL, with average NSE improvements of 5.77%, 7.19%, and 21.70% using 2-year, 1-year, and 6-month target training data, respectively. Moreover, while both

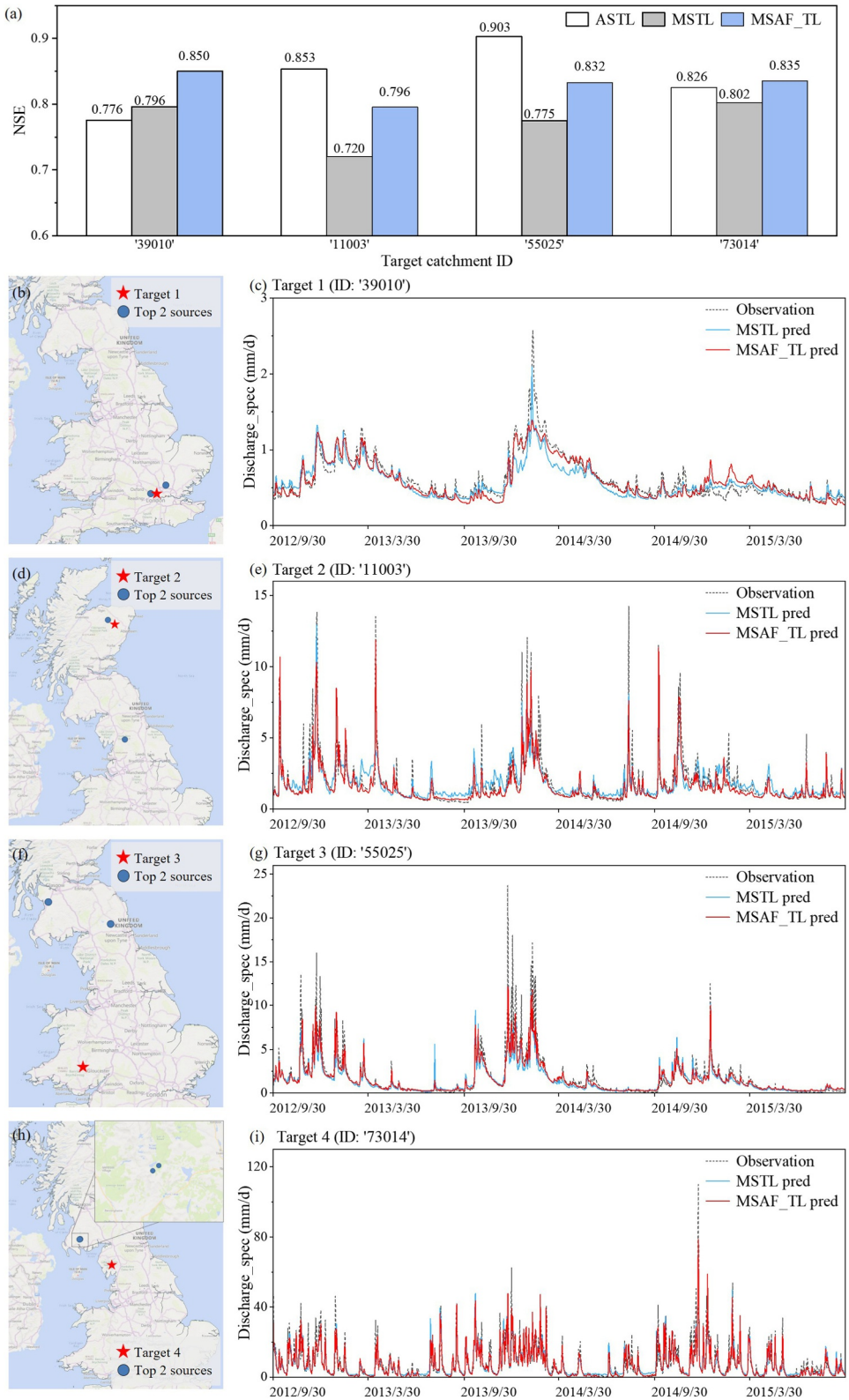


Figure 5. Transfer learning performance across four target catchments using top 2 source catchments. (a) NSE comparison between ASTL, MSTL and MSAF-TL (b, d, f, h) Spatial distribution of target catchments (red stars) and their respective top 2 source catchments (blue circles) (c, e, g, i) Observed and predicted discharge time series for the testing period.

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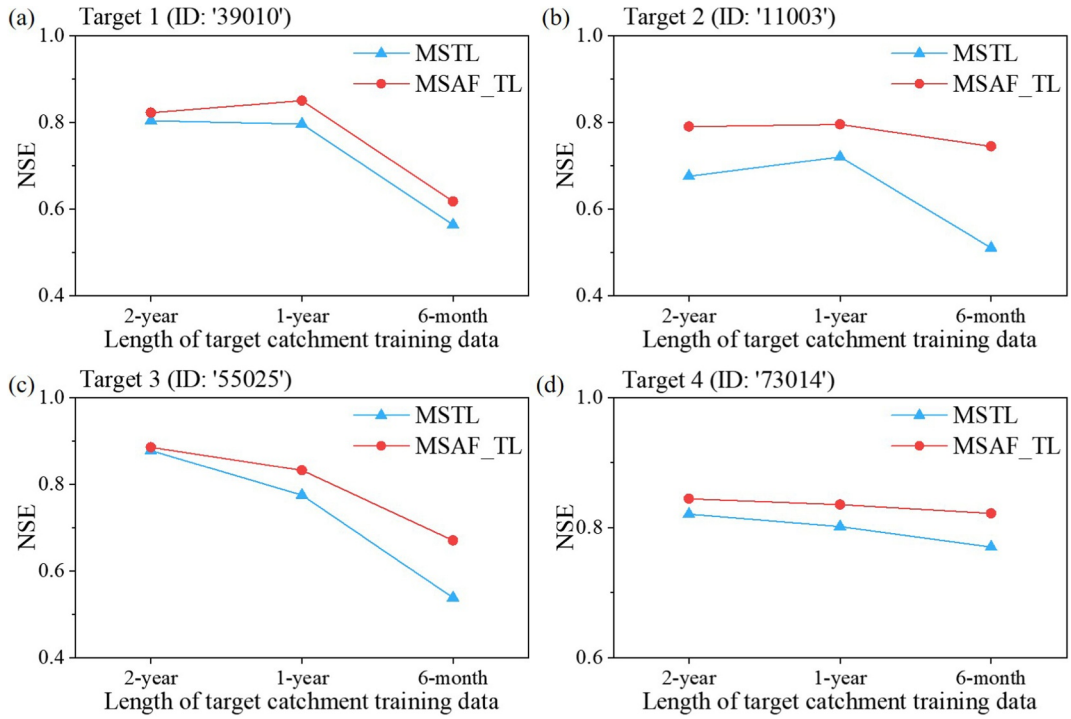


Figure 6. Effect of training data length from target catchment on NSE performance of MSTL and MSAF-TL across four target catchments: (a–d) Target 1–4.

networks show declining performance as target data decreases, MSAF-TL demonstrates greater resilience to data scarcity. For target "39010" (Figure 6a), MSAF-TL yields NSE improvements of 2.25%, 6.78%, and 9.54% over MSTL with the aforementioned three target training data length. The advantage is more pronounced for target "11003", with gains of 17.03%, 10.41%, and 45.94% (Figure 6b). Similar trends are observed for targets "55025" and "73014", with MSAF-TL outperforming MSTL by 0.88%–24.63% and 2.90%–6.68%, respectively (Figures 6c and 6d).

Figure 7 further illustrates model performance from a flow-distribution perspective by presenting FDCs for the same experiments. As the target training data decreases, the simulated FDCs from both MSTL and MSAF-TL gradually deviate from the observed ones; however, MSAF-TL consistently maintains smaller departures. For target "39010", when the target training data is reduced from 2 years (Figure 7a) to 1 year (Figure 7e), MSTL shows an evident mismatch in the medium-to-high flow range, whereas MSAF-TL preserves a closer align with observations. With only 6 months of data (Figure 7i), both models exhibit a noticeable performance drop, which is likely attributable to pronounced seasonality in this catchment and the insufficient representativeness of a 6-month record. For Target "11003", under 6-month training data setting (Figure 7j), MSAF-TL remains better aligned with the observed FDC than MSTL across both medium-to-high flows and low flows. For Targets "55025" and "73014", both methods retain relatively stable FDC shapes as target training data decreases, yet MSAF-TL shows an advantage in the low-flow regime, even with only 6 months of target data (Figures 7k and 7l). The results show that our proposed adaptive fusion mechanism effectively compensates for limited target information, maintaining predictive accuracy even under highly data constrained conditions.

5. Discussion

5.1. Effectiveness of the Proposed MSAF-TL Network

By evaluating the performance of MSAF-TL against MSTL, the results demonstrate that our adaptive fusion strategy is more effective and robust than the conventional input-level fusion strategy when limited sources are

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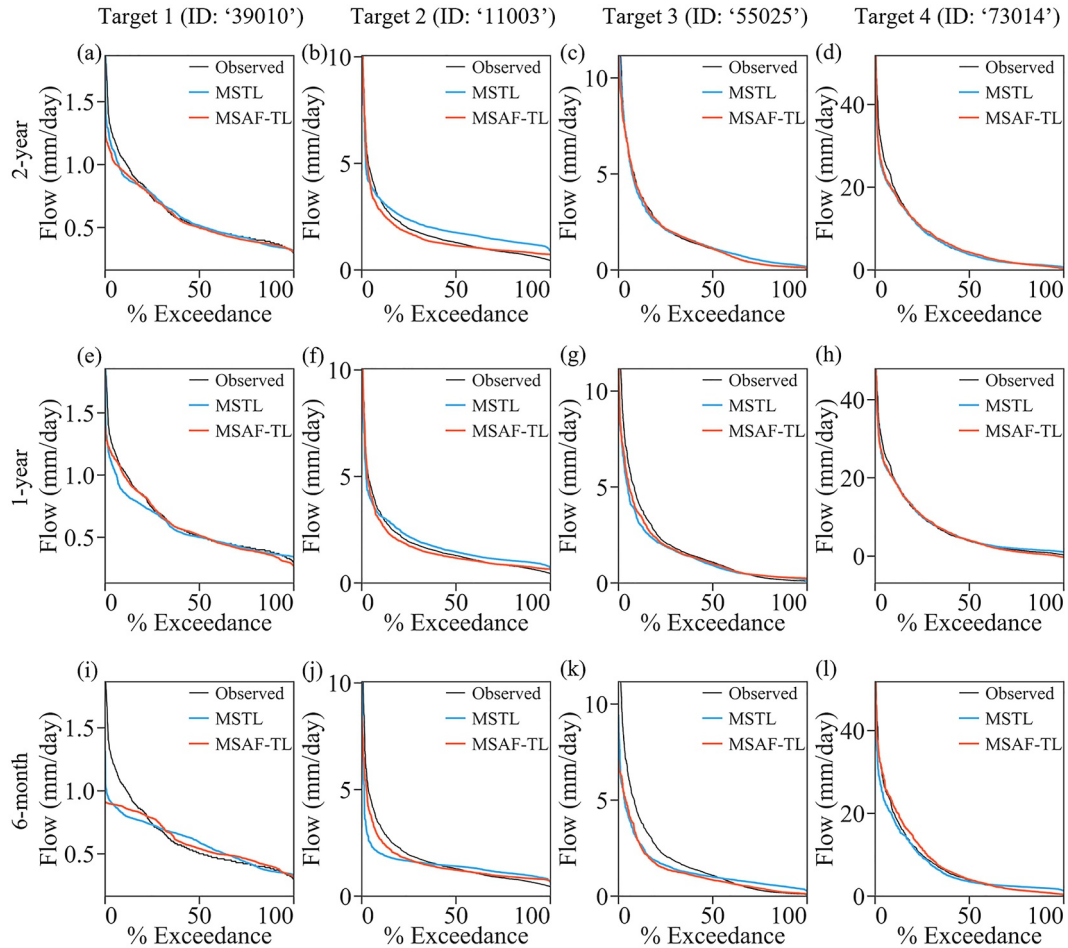


Figure 7. Flow duration curves (FDCs) for the four target catchments comparing observed streamflow with simulations from MSTL and MSAF-TL. The first row (a–d) uses a 2-year target training data set, the second row (e–h) uses a 1-year training data set, and the third row (i–l) uses a 6-month training data set. Columns correspond to the four target catchments (Target 1–4, left to right).

available. This advantage is particularly pronounced for high-flow and low-flow regimes. High flows are not solely controlled by rainfall intensity, but also strongly modulated by catchment-specific conditions, including antecedent soil moisture conditions, topography, and routing effects. In contrast, low flows are more closely linked to subsurface controls, especially groundwater storage and recharge that govern recession behavior and baseflow sustainment. These controls suggest that effective TL should capture two types of knowledge: (a) how rainfall–runoff response dynamics depend on key static catchment attributes, and (b) which response patterns learned from source catchments are similar to the target.

Compared with ASTL, both MSTL and MSAF-TL perform a source selection step, which increases the likelihood that the chosen sources share similar extreme-event responses with the target (knowledge B). Under this setting, simulation performance becomes more sensitive to how efficiently the model can extract rare extreme-event signals from limited samples and correctly associate them with relevant static attributes (knowledge A). However, MSTL relies on simple input-level fusion and a relatively broad fine-tuning strategy, which has been widely explored in previous hydrological TL studies (Khoshkalam et al., 2023). This implementation may limit its ability to fully exploit the source knowledge. In contrast, the MSAF-TL process targets data through multiple independent feature extraction pathways, merging multi-dimensional information at each layer. In hydrology, it is reasonable to assume that different dimensions of features often represent different aspects of hydrological patterns — deeper

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layers capture broader, long-term cycles (e.g., annual cycles), while shallower layers capture more granular details (e.g., daily variations) (Chung et al., 2016). This enables the model to selectively extract and fuse pretrained knowledge at the appropriate scale for floods and droughts. Furthermore, the cross-attention-based mechanism can automatically extract different useful information from each source for the target catchment, more effectively capturing deep dynamic relationships and integrating complementary features between sources (Zhang et al., 2024).

While the current MSAF-TL design uses static attribute similarity to guide the weighting scheme, future work could also incorporate similarity in dynamic variables, for example, by deriving a dynamic similarity embedding from rainfall and historical streamflow windows and integrating it with the static similarity score.

5.2. Influence of Source Catchment Complementarity on TL Performance

Results from Section 4.2 indicate that MSAF-TL exhibits greater performance improvements for Targets 2 and 3 compared to Targets 1 and 4, suggesting that MSAF-TL is particularly effective at extracting complementary information between selected source catchments. For instance, the top 2 source catchments ("39023" and "39019") for Target 1 are geographically close and hydrologically similar—sharing chalky geology, gentle valley slopes, and comparable climate characteristics (e.g., SAARs of 755 and 736 mm). While such similarity is advantageous for each source to conduct direct knowledge transfer to Target 1, it inevitably limits the complementary knowledge provided by the two source catchments. The situation for Target "73014" is more pronounced, where the two sources are located closer to each other.

In contrast, Target 2 benefits from source catchments ("27047" and "9005") that differ significantly in hydrogeology and climate. Catchment "27047" consists 100% of moderately permeable Carboniferous limestone bedrock with a SAAR of 1,729 mm, whereas catchment "9005" features 91.43% of very low permeability metamorphic basement with moorland and forestry, and a SAAR of 1,071 mm. This high degree of complementarity allows MSAF-TL to achieve its maximum advantage, with an NSE improvement of up to 10.41%. A similar trend holds for Target 3. These findings highlight the importance of considering not only source–target similarity but also inter-source complementarity when selecting sources for TL (Afridi et al., 2018).

Additionally, results from Section 4.3 show that NSE performance improvements under reduced target data availability (from 2-year to 6-month) are also more pronounced for Targets 2 and 3. The finding suggests that the complementary knowledge captured by MSAF-TL plays an increasingly critical role in more data-scarce situations.

Future work should further quantify source catchment complementarity by defining a metric for the additional, non-redundant information provided by each source relative to the others. For example, SHapley Additive exPlanations (SHAP)-based analysis could serve as one possible approach. This would help justify source selection and clarify the contributions of different sources.

5.3. Limitations and Future Work

One inherent limitation shared by all tuning-based TL methods, including MSAF-TL, is the reliance on labeled data from the target catchment, such as streamflow records. This restricts their direct application to completely ungauged catchments. To address this, future work could explore alternative strategies, such as attribute similarity-based regionalization, unsupervised domain adaptation, and weak supervision by synthetic streamflow, to reduce or remove dependence on target labels (Farahani et al., 2021).

Another question worth exploring is how to determine the optimal number and data length of source catchments for a specific target catchment. This study has not yet examined the impact of the training data length of the source catchment. We speculate that for different target catchments, there may be an optimal (or saturation) point in terms of the number and duration of source data. Beyond this point, additional data may introduce redundancy or noise, which could degrade performance unless handled with the more sophisticated TL architectures. Such a saturation point may shift due to several factors. The first factor is the hydrological

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dissimilarity between source and target catchments. In cross-continental TL applications, it is likely that a larger number of sources will be needed to capture the broader variability in hydrological patterns. The second factor is the data quality of both source and target catchments. When dealing with lower-quality data sets characterized by missing values, measurement errors, or shorter time series, additional sources are then required to ensure reliable knowledge transfer. By identifying this saturation point, an optimal number of sources embedded in the MSAF-TL network can raise the upper performance bound brought by the ASTL network. The need for finding the optimal point coincides with the statement of a previous study that there is a significant research gap in addressing the trade-off between the advantages of large data sets and the requirement for specific target catchment applicability (Nai et al., 2024).

In addition, while this study primarily focuses on data-scarce target settings, where TL is most needed and the transfer mechanism is most informative, the effect of using longer target training records also deserves investigation. Extending the fine-tuning experiments to longer target data lengths, such as 3–5 years, would further help reveal how TL gains evolve as target information increases and whether diminishing returns emerge.

Additionally, beyond pure data-driven TL approaches, future research should consider incorporating physical mechanisms to assist the TL process for target catchments. For example, synthetic streamflow data from distributed hydrological models and supplemental extreme event samples could be used to enhance TL model performance (Kang et al., 2021; Slater et al., 2025). Physical conservation principles (e.g., water and energy balance) can also be used as constraints within the neural network architecture, enabling the model to maintain physical consistency while adapting to new target domains. Such physics-informed approaches can also enhance both the interpretability and generalization capability of TL models.

6. Conclusion

The objectives of this study are twofold. One is to achieve more accurate and stable TL performance in regions with limited source catchments, and the other is to reduce the data requirements, and thus computational demands, in large-scale hydrological TL without compromising performance through source pre-selection. This study presents a novel MSAF-TL network to enhance streamflow forecasting in data-scarce catchments. By incorporating flow regime similarity analysis for source catchment pre-selection and employing a cross-attention-based adaptive fusion strategy, MSAF-TL consistently outperforms conventional fine-tuning-based TL approaches (MSTL). Evaluated across four target catchments with diverse climatic and geological characteristics, MSAF-TL achieves NSE improvements ranging from 4.19% to 10.41% compared to MSTL using only 1 year of the target training data. These performance gains become even more pronounced as the length of the target data is reduced from 2 years to 6 months, particularly under high-flow and low-flow conditions, highlighting the model's resilience in data-scarce scenarios. The advantage stems from MSAF-TL's multi-level AFM, which can selectively extract and fuse pretrained knowledge at the appropriate fine-to-broad scale for high-flow and baseflow simulation. The study could be further extended to completely ungauged catchments by incorporating unsupervised DL strategies. In addition, more systematic tests across different target training lengths, as well as varying the number and combinations of source catchments, may help identify data efficient source target configurations that retain the most complementary information while minimizing data requirements. Overall, MSAF-TL exhibits a promising potential for improving hydrological prediction in data-scarce regions.

Appendix A

Figure A1

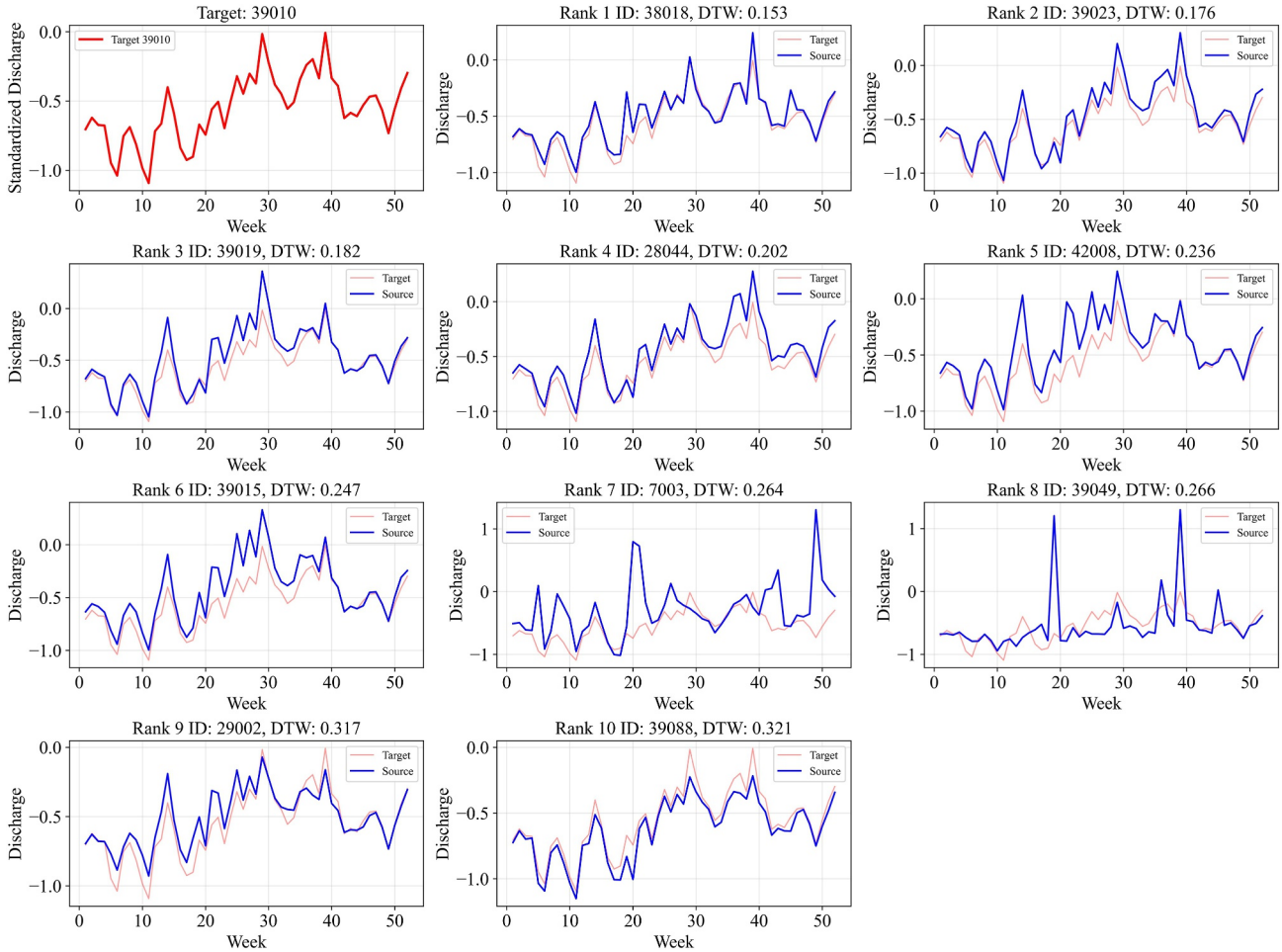


Figure A1. Comparison between the weekly hydrographs of the top 10 source catchments and the target catchment (ID: 39010), along with their corresponding Dynamic Time Warping (DTW) distances.

Appendix B

The optimal values of each hyperparameter for every experiment are listed in Tables B1–B6 below.

Table B1
Optimal Hyperparameters Within Given Ranges of NTL Network

Target	bs (32, 64)	seq (90, 180, 365)	dr (0.4, 0.6)	lr (10^{-3}, 10^{-4})
"39010"	32	365	0.4	10^{-3}
"11003"	32	365	0.6	10^{-3}
"55025"	64	365	0.6	10^{-3}
"73014"	32	365	0.6	10^{-3}

Note. bs, batch size; seq, sequence length; dr, output dropout; lr, initial learning rate.

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Table B2
Optimal Hyperparameters Within Given Ranges of ASTL Network

Pre-training stage					Fine-tuning stage				
Sources	bs (32, 64)	seq (90, 180, 365)	dr (0.4, 0.6)	lr (10^{-3}, 10^{-4})	Target	bs (32, 64)	dr (0.4, 0.6)	lr (10^{-3}, 10^{-4})	ft (A, B, C)
All source catchments except target	32	365	0.4	10^{-4}	"39010"	32	0.4	10^{-4}	C
	32	365	0.4	10^{-4}	"11003"	64	0.6	10^{-4}	B
	32	365	0.4	10^{-4}	"55025"	32	0.4	10^{-4}	C
	32	365	0.4	10^{-4}	"73014"	32	0.6	10^{-4}	C

Note. bs, batch size; seq, sequence length; dr, output dropout; lr, initial learning rate; ft, fine-tuning strategy.

Table B3
Optimal Hyperparameters Within Given Ranges of Pre-Training Experiments of MSTL Network

		Pre-training stage			
Target	Sources	bs (32, 64)	seq (90, 180, 365)	dr (0.4, 0.6)	lr (10^{-3}, 10^{-4})
"39010"	Top 2	32	365	0.4	10^{-3}
	Top 6	64	365	0.4	10^{-3}
	Top 10	64	365	0.4	10^{-3}
"11003"	Top 2	32	365	0.4	10^{-3}
"55025"		32	365	0.4	10^{-3}
"73014"		32	365	0.6	10^{-3}
"39010"	Top 2	32	365	0.4	10^{-3}
"11003"		32	365	0.4	10^{-3}
"55025"		32	365	0.4	10^{-3}
"73014"		32	365	0.6	10^{-3}

Note. bs, batch size; seq, sequence length; dr, output dropout; lr, initial learning rate.

Table B4
Optimal Hyperparameters Within Given Ranges of Fine-Tuning Experiments of MSTL Network

		Fine-tuning stage			
Target	Target training data length	bs (32, 64)	dr ([0.4, 0.6])	lr ([1e-3, 1e-4])	ft (A, B, C)
"39010"	1-year	32	0.6	10^{-3}	C
		64	0.4	10^{-3}	C
		64	0.4	10^{-3}	C
"11003"	1-year	32	0.4	10^{-3}	C
"55025"		64	0.4	10^{-3}	B
"73014"		32	0.4	10^{-3}	C
"39010"	2-year	32	0.4	10^{-3}	C
	6-month	64	0.6	10^{-3}	C
"11003"	2-year	32	0.4	10^{-3}	C
	6-month	32	0.4	10^{-3}	C
"55025"	2-year	64	0.6	10^{-3}	C
	6-month	64	0.4	10^{-4}	C

Table B4
Continued

Target	Target training data length	Fine-tuning stage bs (32, 64)	dr ([0.4, 0.6])	lr ([1e-3, 1e-4])	ft (A, B, C)
"73014"	2-year	64	0.4	10^{-3}	C
	6-month	32	0.6	10^{-3}	C

Note. bs, batch size; dr, output dropout; lr, initial learning rate; ft, fine-tuning strategy.

Table B5
Optimal Hyperparameters Within Given Ranges of Pre-Training Experiments of MSAF-TL Network

Target	Sources	Pre-training stage bs (32, 64)	seq (90, 180, 365)	dr (0.4, 0.6)	lr (10^{-3}, 10^{-4})
"39010"	Source 1: 38018	64	365	0.6	10^{-3}
	Source 2: 39023	64	365	0.6	10^{-3}
	Source 3: 39019	64	365	0.4	10^{-3}
	Source 4: 28044	32	365	0.4	10^{-4}
	Source 5: 42008	32	365	0.6	10^{-3}
	Source 6: 39015	32	365	0.6	10^{-3}
	Source 7: 7003	64	365	0.6	10^{-3}
	Source 8: 39049	64	365	0.6	10^{-3}
	Source 9: 29002	64	365	0.4	10^{-3}
	Source 10: 39088	32	365	0.4	10^{-3}
"11003"	Source 1: 27047	32	365	0.6	10^{-3}
	Source 2: 9005	32	365	0.4	10^{-3}
"55025"	Source 1: 83013	32	365	0.6	10^{-3}
	Source 2: 23008	32	365	0.4	10^{-3}
"73014"	Source 1: 80004	32	365	0.6	10^{-3}
	Source 2: 80003	32	365	0.6	10^{-3}

Note. bs, batch size; seq, sequence length; dr, output dropout; lr, initial learning rate.

Table B6
Optimal Hyperparameters Within Given Ranges of Fine-Tuning Experiments of MSAF-TL Network

Target	Target training data length	Fine-tuning stage bs (32, 64)	dr (0.4, 0.6)	lr (10^{-3}, 10^{-4})	ft (A, B, C)	sdr (0.4, 0.6)	cr (10, 20)	Temp (1.5, 2.0)
"39010"	1-year	32	0.4	10^{-3}	B	0.6	10	1.5
		32	0.4	10^{-3}	B	0.4	20	1.5
		32	0.4	10^{-3}	B	0.4	20	1.5
"11003"	1-year	32	0.4	10^{-3}	C	0.6	20	1.5
"55025"		64	0.6	10^{-3}	C	0.4	20	1.5
"73014"		64	0.6	10^{-3}	C	0.4	20	1.5
"39010"	2-year	32	0.4	10^{-3}	C	0.4	20	1.5
	6-month	64	0.4	10^{-4}	B	0.4	20	2

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Table B6
Continued

		Fine-tuning stage						
Target	Target training data length	bs (32, 64)	dr (0.4, 0.6)	lr (10^{-3}, 10^{-4})	ft (A, B, C)	sdr (0.4, 0.6)	cr (10, 20)	Temp (1.5, 2.0)
"11003"	2-year	64	0.6	10^{-3}	C	0.6	10	2
	6-month	32	0.6	10^{-3}	C	0.4	20	1.5
"55025"	2-year	32	0.6	10^{-3}	C	0.4	20	1.5
	6-month	32	0.4	10^{-3}	C	0.4	20	1.5
"73014"	2-year	64	0.4	10^{-3}	C	0.4	20	1.5
	6-month	32	0.4	10^{-3}	B	0.4	20	1.5

Note. bs, batch size; dr, output dropout; lr, initial learning rate; ft, fine-tuning strategy; sdr, static similarity dropout; cr, clamp range; temp—temperature.

Appendix C

All metrics for the training and testing phases are provided in the tables below. Tables C1–C3 are the complete results for experiments in main text Section 4.2. Tables C4 and C5 are the complete results for experiments in main text Section 4.3.

Table C1
All Metrics for the Training and Testing Phases in the ASTL Network Experiments Evaluating Across Diverse Target Catchments

		ASTL				
Target	Phase	NSE	KGE	FHV	FMS	FLV
"39010"	Training	0.703	0.819	9.912	11.101	−9.043
	Testing	0.776	0.797	−10.199	−19.292	17.519
"11003"	Training	0.811	0.873	−12.298	0.090	−7.035
	Testing	0.853	0.882	−9.642	−8.665	−31.682
"55025"	Training	0.928	0.929	−4.994	−13.123	64.199
	Testing	0.903	0.862	−12.490	−13.086	72.791
"73014"	Training	0.808	0.830	−9.810	−2.728	12.876
	Testing	0.826	0.865	−10.007	−2.372	−87.495

Table C2
All Metrics for the Training and Testing Phases in the MSTL Network Experiments Evaluating Across Diverse Target Catchments

		MSTL				
Target	Phase	NSE	KGE	FHV	FMS	FLV
"39010"	Training	0.511	0.726	−3.170	−21.006	71.962
	Testing	0.796	0.740	−21.548	−16.686	58.313
"11003"	Training	0.584	0.635	−26.770	−6.112	24.893
	Testing	0.720	0.729	−19.271	−12.875	17.093
"55025"	Training	0.791	0.741	−19.755	−19.063	−64.545
	Testing	0.775	0.641	−32.403	−12.691	−38.174
"73014"	Training	0.771	0.801	−7.816	−5.497	53.203
	Testing	0.802	0.827	−15.002	−7.555	74.166

Table C3
All Metrics for the Training and Testing Phases in the MSAF-TL Network Experiments Evaluating Across Diverse Target Catchments

		MSAF-TL				
Target	Phase	NSE	KGE	FHV	FMS	FLV
"39010"	Training	0.766	0.856	9.009	−1.943	67.074
	Testing	0.850	0.852	−25.611	4.701	−12.095
"11003"	Training	0.713	0.777	−16.724	−2.895	41.311
	Testing	0.796	0.820	−10.501	−12.491	59.578
"55025"	Training	0.860	0.818	−20.088	−17.521	35.014
	Testing	0.832	0.706	−31.150	−10.238	56.105
"73014"	Training	0.829	0.857	−4.957	7.144	−947.078
	Testing	0.835	0.854	−14.708	12.910	−634.064

Table C4
All Metrics for the Training and Testing Phases in the MSTL Network Experiments Evaluating Across Various Data Lengths From Target Catchments

		MSTL					
Selected targets ID	Target training data length	Phase	NSE	KGE	FHV	FMS	FLV
"39010"	2-year	Training	0.593	0.808	10.189	−10.760	66.774
		Testing	0.804	0.805	−20.180	−3.967	37.890
	6-month	Training	−0.114	0.632	−5.044	3.991	63.933
		Testing	0.564	0.541	−41.630	−26.602	43.334
"11003"	2-year	Training	0.636	0.641	−25.244	−29.275	43.632
		Testing	0.675	0.648	−16.466	−36.146	28.128
	6-month	Training	0.340	0.373	−37.859	−58.418	27.898
		Testing	0.510	0.451	−38.515	−57.350	14.115
"55025"	2-year	Training	0.873	0.833	−6.527	−24.832	39.188
		Testing	0.878	0.837	−18.516	−23.348	26.273
	6-month	Training	0.526	0.575	−32.441	−46.880	38.520
		Testing	0.538	0.451	−46.416	−50.007	37.600
"73014"	2-year	Training	0.785	0.787	−8.912	−2.409	22.831
		Testing	0.821	0.799	−19.165	0.135	65.400
	6-month	Training	0.710	0.686	−26.207	−15.857	61.570
		Testing	0.770	0.754	−23.996	−17.194	73.420

Table C5
All Metrics for the Training and Testing Phases in the MSAF-TL Network Experiments Evaluating Across Various Data Lengths From Target Catchments

		MSAF-TL					
Selected targets ID	Target training data length	Phase	NSE	KGE	FHV	FMS	FLV
"39010"	2-year	Training	0.760	0.872	−1.294	−8.253	64.512
		Testing	0.822	0.762	−31.099	0.315	24.395
	6-month	Training	0.640	0.752	−15.053	−27.677	46.963
		Testing	0.617	0.568	−47.555	−20.587	−3.701

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Table C5
Continued

		MSAF-TL					
Selected targets ID	Target training data length	Phase	NSE	KGE	FHV	FMS	FLV
"11003"	2-year	Training	0.745	0.793	−12.939	−6.387	54.685
		Testing	0.790	0.796	−10.923	−21.629	68.754
	6-month	Training	0.665	0.599	−33.190	−18.854	53.741
		Testing	0.745	0.654	−30.655	−26.212	60.293
"55025"	2-year	Training	0.918	0.907	−14.950	1.610	−125.099
		Testing	0.885	0.831	−25.528	1.484	−71.322
	6-month	Training	0.740	0.575	−41.243	−35.521	21.560
		Testing	0.670	0.467	−50.221	−33.249	32.286
"73014"	2-year	Training	0.825	0.838	−4.910	−7.966	−44.748
		Testing	0.844	0.833	−16.434	−2.894	29.384
	6-month	Training	0.710	0.686	−26.207	−15.857	61.570
		Testing	0.822	0.888	−14.030	10.082	36.189

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The open-source code Neuralhydrology [Software] was used in this work (Kratzert et al., 2022), available at https://github.com/neuralhydrology/neuralhydrology. The proposed MSAF-TL network code is available at https://github.com/YuxuanGaoYG/MSAF-TL. The CAMELS-GB [Dataset] was used in this work (Coxon et al., 2020), available via the UK Centre for Ecology & Hydrology Environmental Information Data Centre (https://doi.org/10.5285/8344e4f3-d2ea-44f5-8afa-86d2987543a9).

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